

Age Effect Analysis

METIMMOX-1 Economic Evaluation

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Table of contents

1	Introduction	1
2	Age Distribution in the Sample	3
2.1	Overall Age Statistics	3
2.2	Age by Treatment Arm	3
2.3	Age by Biomarker Status	3
3	CRP Biomarker Strategy	4
3.1	Baseline Characteristics	4
3.2	Life-Years Comparison	4
3.3	Overall Survival Curves	5
3.4	Progression-Free Survival Curves	5
4	TLR Biomarker Strategy	6
4.1	Baseline Characteristics	6
4.2	Life-Years Comparison	6
4.3	Overall Survival Curves	7
4.4	Progression-Free Survival Curves	7
5	TMB/BRAF Biomarker Strategy	8
5.1	Baseline Characteristics	8
5.2	Life-Years Comparison	8
5.3	Overall Survival Curves	9
5.4	Progression-Free Survival Curves	9
6	Cost-Effectiveness Impact	10
6.1	QALYs Comparison	10
6.2	Costs Comparison	10
6.3	ICERs Comparison	10
7	Summary	11
7.1	Key Findings	11
7.2	Interpretation	11

1 Introduction

This technical document investigates the impact of age heterogeneity on survival model predictions and cost-effectiveness results in the METIMMOX-1 economic evaluation. All survival

models in this analysis include Age as a covariate, meaning patient-level age variation directly influences predicted outcomes.

Analysis approach:

1. **Base case:** Use actual patient ages from the dataset (standard analysis)
2. **Same-age scenario:** Replace all patient ages with the population mean age

By comparing these two approaches, we can quantify how much of the predicted survival differences across biomarker subgroups is attributable to age imbalances versus true biomarker-treatment effects.

Important note: The same-age dataset is created and used exclusively within this document to prevent contamination of other analyses in the codebase.

2 Age Distribution in the Sample

This section presents the age distribution across the study population, stratified by treatment arm and biomarker status.

2.1 Overall Age Statistics

Table 1: Overall age statistics in the study population.

Statistic	Value
Mean	64
SD	10
Median	64
Min	38
Max	80
N	65

The population mean age of **64** years is used for the same-age scenario throughout this analysis.

2.2 Age by Treatment Arm

Table 2: Age distribution by treatment arm.

Treatment Arm	N	Mean Age	SD
Control arm	29	65.0	10.8
Experimental arm	36	63.2	9.4

2.3 Age by Biomarker Status

Table 3: Age distribution by biomarker status. Mean Age values are highlighted as they represent the key variable of interest in this analysis.

	Biomarker	Status	N	Mean Age	SD
crp	CRP	Negative	43	63.8	10.0
crp1	CRP	Positive	22	64.3	10.3
tlr	TLR	Negative	24	62.2	10.2
tlr1	TLR	Positive	41	65.0	9.9
tmb_braf	TMB/BRAF	Negative	36	63.5	9.8
tmb_braf1	TMB/BRAF	Positive	29	64.6	10.4

3 CRP Biomarker Strategy

C-reactive protein (CRP) with a cutoff of <5 g/L for biomarker-positive status. **Biomarker prevalence: 33.8%.**

3.1 Baseline Characteristics

Table 4: CRP strategy: Baseline characteristics by biomarker status and treatment arm.

Variable	Biomarker-		Biomarker+	
	Biomarker- Control	Biomarker- Experimental	Biomarker+ Control	Biomarker+ Experimental
N	24	19	5	17
Age, mean (SD)	66.3 (10.2)	60.7 (9)	58.6 (12.8)	65.9 (9.2)
Sex, n (%) Male	15 (62.5%)	10 (52.6%)	2 (40%)	8 (47.1%)

3.2 Life-Years Comparison

Table 5: CRP strategy: Overall survival life-years. Base case uses actual ages; Same-Age uses population mean age for all patients.

Model	Ctrl Base	Ctrl Same-Age	Ctrl Delta	Strat Base	Strat Same-Age	Strat Delta
Model A	2.17	2.18	0.01	2.09	2.09	0.00
Model B	2.23	2.25	0.02	2.13	2.13	-0.01
Model C	2.46	2.46	0.00	2.24	2.24	0.01

Table 6: CRP strategy: Progression-free survival life-years. Base case uses actual ages; Same-Age uses population mean age for all patients.

Model	Ctrl Base	Ctrl Same-Age	Ctrl Delta	Strat Base	Strat Same-Age	Strat Delta
Model A	0.90	0.90	0.00	1.20	1.20	0.00
Model B	0.95	0.95	0.00	1.21	1.21	0.00
Model C	1.07	1.04	-0.02	1.33	1.34	0.01

3.3 Overall Survival Curves

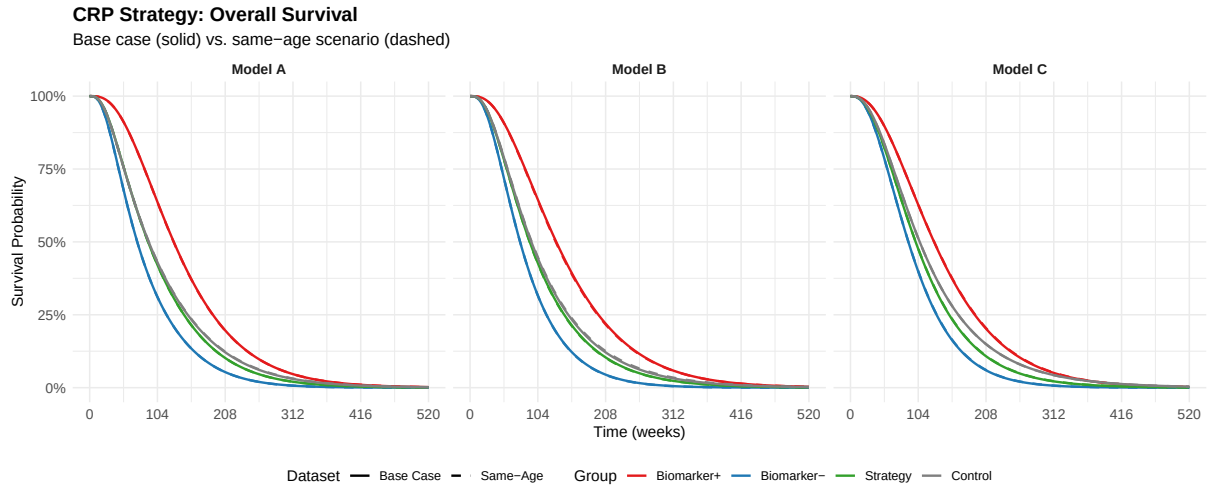


Figure 1: CRP strategy (prevalence: 33.8%) overall survival comparison. Solid lines = base case (actual ages), dashed lines = same-age scenario (all patients at mean age 64 years).

3.4 Progression-Free Survival Curves

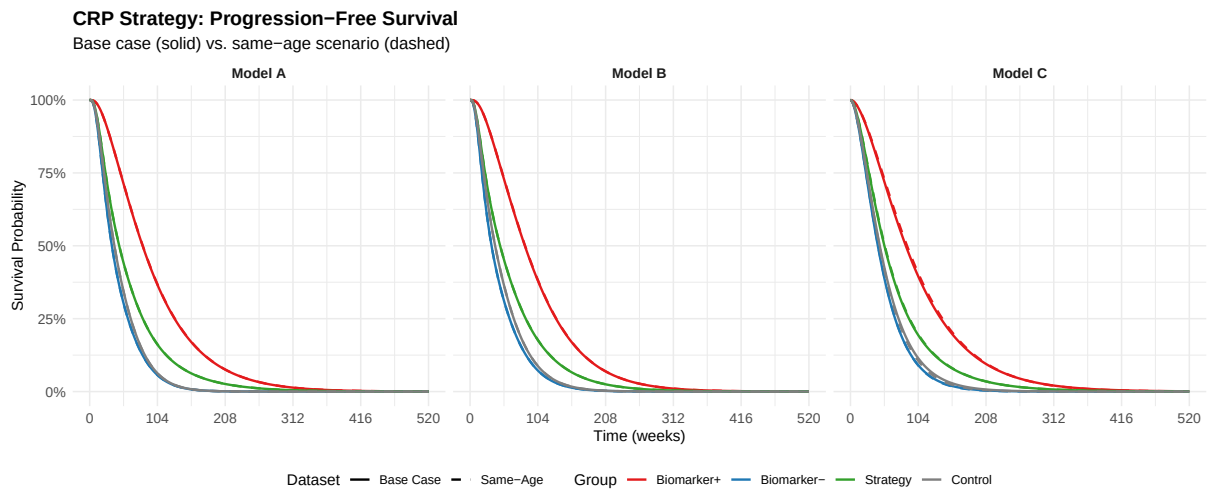


Figure 2: CRP strategy (prevalence: 33.8%) progression-free survival comparison. Solid lines = base case (actual ages), dashed lines = same-age scenario (all patients at mean age 64 years).

4 TLR Biomarker Strategy

Tumor lesion reduction (TLR) with a cutoff of $\geq 10\%$ for biomarker-positive status. **Biomarker prevalence: 64.7%.**

4.1 Baseline Characteristics

Table 7: TLR strategy: Baseline characteristics by biomarker status and treatment arm.

Variable	Biomarker-		Biomarker+	
	Biomarker- Control	Biomarker- Experimental	Biomarker+ Control	Biomarker+ Experimental
N	7	17	22	19
Age, mean (SD)	68.7 (8.6)	59.6 (9.8)	63.8 (11.4)	66.4 (7.8)
Sex, n (%) Male	5 (71.4%)	8 (47.1%)	12 (54.5%)	10 (52.6%)

4.2 Life-Years Comparison

Table 8: TLR strategy: Overall survival life-years. Base case uses actual ages; Same-Age uses population mean age for all patients.

Model	Ctrl Base	Ctrl Same-Age	Ctrl Delta	Strat Base	Strat Same-Age	Strat Delta
Model A	2.17	2.18	0.01	1.99	1.99	0.00
Model B	2.18	2.19	0.00	1.99	1.98	-0.01
Model C	2.01	2.01	0.00	2.10	2.09	0.00

Table 9: TLR strategy: Progression-free survival life-years. Base case uses actual ages; Same-Age uses population mean age for all patients.

Model	Ctrl Base	Ctrl Same-Age	Ctrl Delta	Strat Base	Strat Same-Age	Strat Delta
Model A	0.90	0.90	0	1.19	1.20	0
Model B	1.14	1.14	0	1.22	1.22	0
Model C	0.96	0.96	0	1.40	1.40	0

4.3 Overall Survival Curves

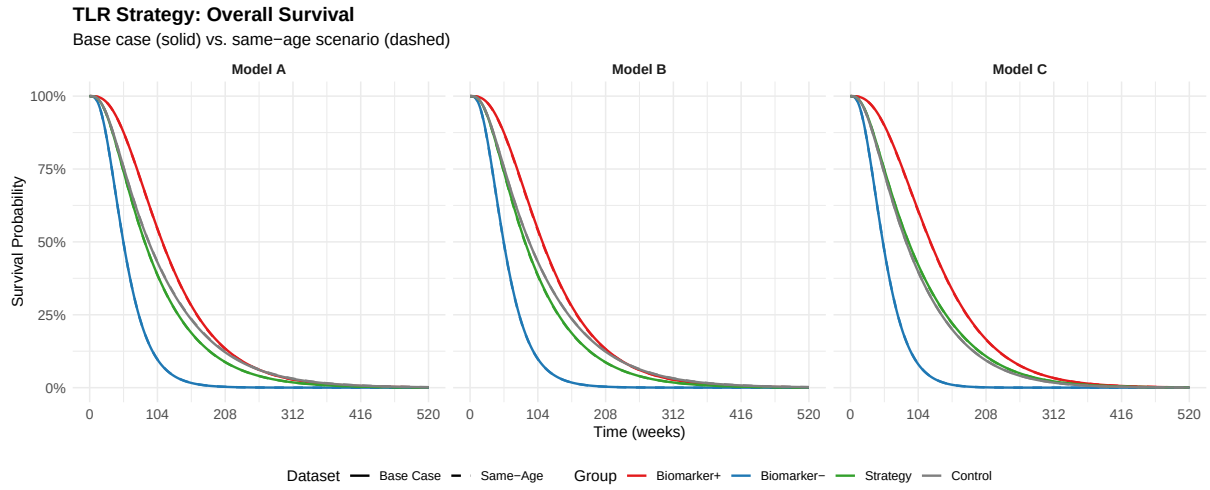


Figure 3: TLR strategy (prevalence: 64.7%) overall survival comparison. Solid lines = base case (actual ages), dashed lines = same-age scenario (all patients at mean age 64 years).

4.4 Progression-Free Survival Curves

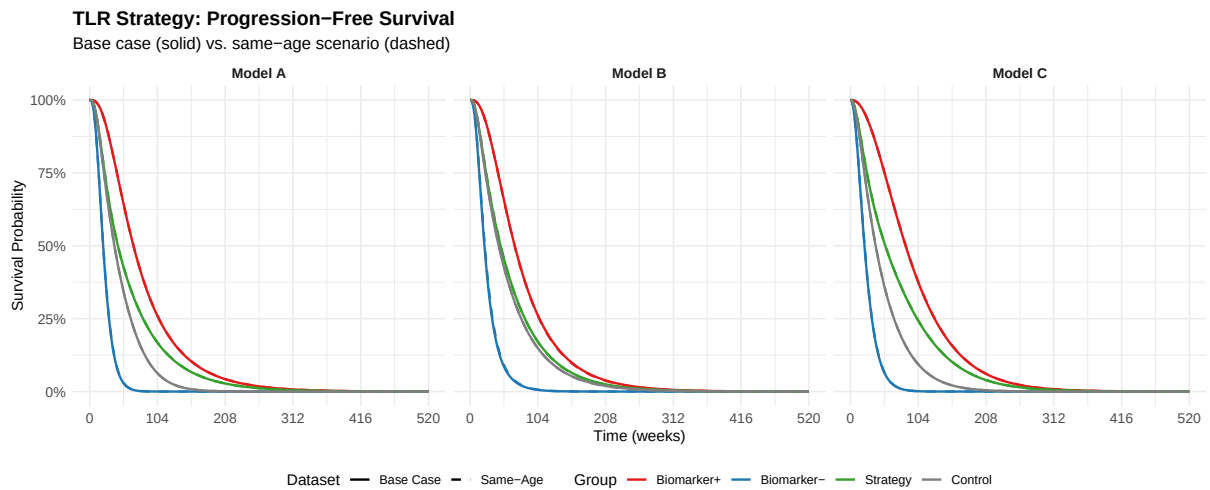


Figure 4: TLR strategy (prevalence: 64.7%) progression-free survival comparison. Solid lines = base case (actual ages), dashed lines = same-age scenario (all patients at mean age 64 years).

5 TMB/BRAF Biomarker Strategy

Combined genomic biomarker: either tumor mutational burden (TMB) ≥ 9 mutations/MB or BRAF V600 mutation positive. **Biomarker prevalence: 44.9%.**

5.1 Baseline Characteristics

Table 10: TMB/BRAF strategy: Baseline characteristics by biomarker status and treatment arm.

Variable	Biomarker-		Biomarker+	
	Biomarker- Control	Biomarker- Experimental	Biomarker+ Control	Biomarker+ Experimental
N	16	20	13	16
Age, mean (SD)	64.2 (9.5)	63 (10.3)	65.9 (12.6)	63.4 (8.4)
Sex, n (%) Male	9 (56.2%)	11 (55%)	8 (61.5%)	7 (43.8%)

5.2 Life-Years Comparison

Table 11: TMB/BRAF strategy: Overall survival life-years. Base case uses actual ages; Same-Age uses population mean age for all patients.

Model	Ctrl Base	Ctrl Same-Age	Ctrl Delta	Strat Base	Strat Same-Age	Strat Delta
Model A	2.17	2.18	0.01	2.20	2.20	0.00
Model B	2.14	2.14	0.00	2.23	2.24	0.01
Model C	2.19	2.20	0.00	2.36	2.36	0.00

Table 12: TMB/BRAF strategy: Progression-free survival life-years. Base case uses actual ages; Same-Age uses population mean age for all patients.

Model	Ctrl Base	Ctrl Same-Age	Ctrl Delta	Strat Base	Strat Same-Age	Strat Delta
Model A	0.90	0.90	0.00	1.15	1.15	0.00
Model B	1.11	1.12	0.01	1.24	1.23	0.00
Model C	1.04	1.02	-0.02	1.41	1.42	0.01

5.3 Overall Survival Curves

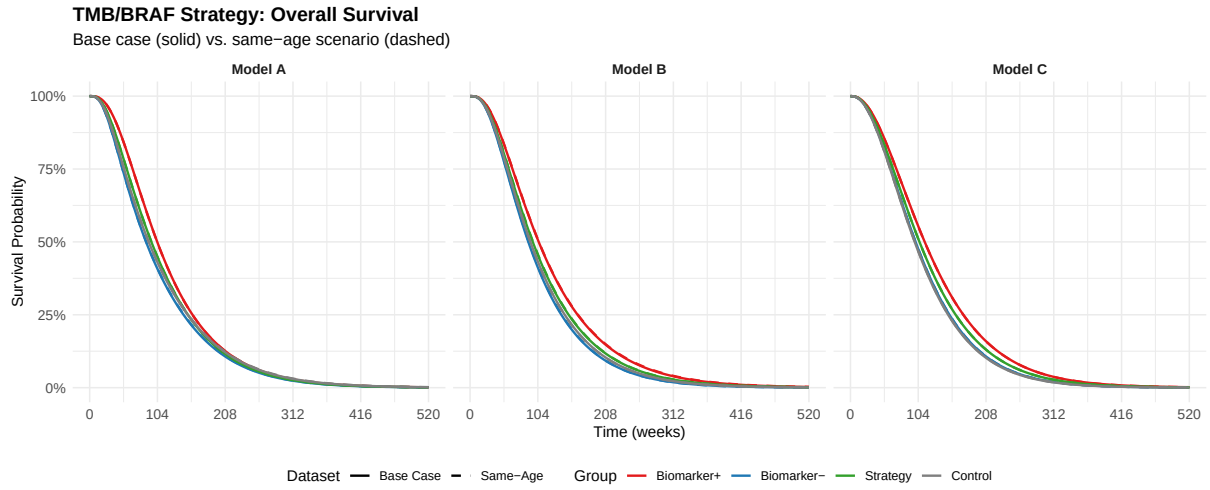


Figure 5: TMB/BRAF strategy (prevalence: 44.9%) overall survival comparison. Solid lines = base case (actual ages), dashed lines = same-age scenario (all patients at mean age 64 years).

5.4 Progression-Free Survival Curves

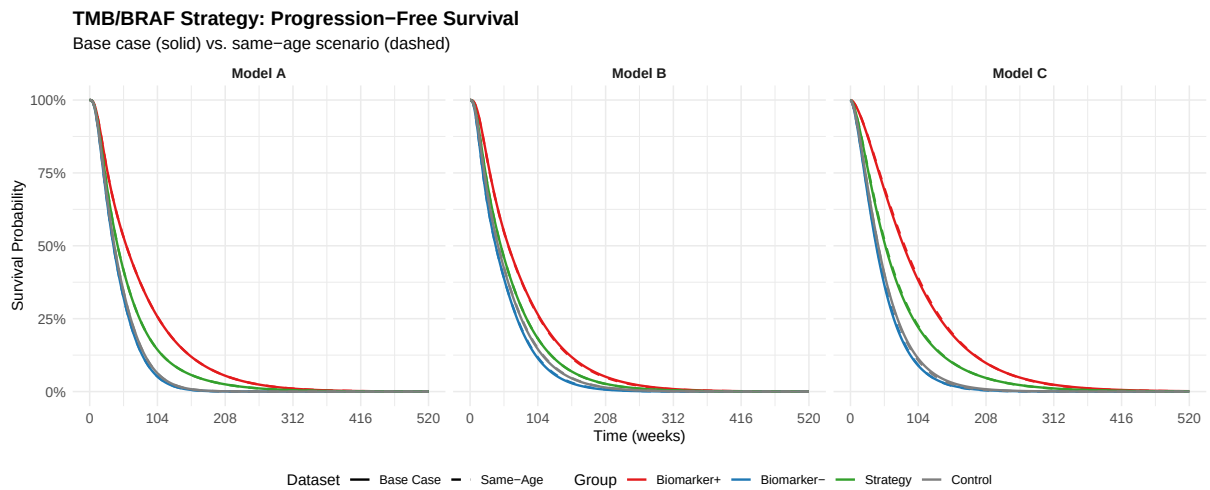


Figure 6: TMB/BRAF strategy (prevalence: 44.9%) progression-free survival comparison. Solid lines = base case (actual ages), dashed lines = same-age scenario (all patients at mean age 64 years).

6 Cost-Effectiveness Impact

This section compares cost-effectiveness results between the base case (actual patient ages) and the same-age scenario (all patients at population mean age). Results use Model A (joint model with all biomarker interactions) for consistency with the primary analysis.

6.1 QALYs Comparison

Table 13: QALYs comparison between base case and same-age scenario (Model A).

Strategy	Base Case	Same-Age	Delta
Control	1.844	1.851	0.007
CRP	1.784	1.783	-0.002
TLR	1.704	1.701	-0.002
TMB/BRAF	1.870	1.873	0.003

6.2 Costs Comparison

Table 14: Costs comparison (EUR) between base case and same-age scenario (Model A).

Strategy	Base Case	Same-Age	Delta
Control	20,894	20,892	-2
CRP	54,887	54,897	9
TLR	84,838	84,835	-2
TMB/BRAF	62,173	62,151	-22

6.3 ICERs Comparison

Note: Negative ICERs may occur when a strategy is dominated (higher costs and lower QALYs) or dominates (lower costs and higher QALYs). While negative ICERs are not conceptually meaningful for decision-making, they are presented here for numerical comparison between the base case and same-age scenarios.

Table 15: ICER comparison (EUR/QALY) between base case and same-age scenario (Model A).

Strategy	ICER Base Case	ICER Same-Age	Delta ICER
CRP vs Control	-567,814	-495,587	72,227
TLR vs Control	-454,913	-425,878	29,035
TMB/BRAF vs Control	1,635,536	1,955,186	319,650

7 Summary

This analysis examined the impact of age heterogeneity on survival predictions and cost-effectiveness results by comparing the base case (actual patient ages) with a same-age scenario where all patients were assigned the population mean age of **64** years.

7.1 Key Findings

1. **Age imbalances across biomarker subgroups:** The mean age varied by up to 1.8 years from the population mean across biomarker subgroups.
2. **Impact on survival predictions:** Standardizing age affected OS life-year predictions by up to 0.01 years, indicating the magnitude of age-related confounding.
3. **Cost-effectiveness implications:** The same-age scenario shows how age differences between biomarker subgroups can influence ICER estimates independently of biomarker-treatment effects.

7.2 Interpretation

When biomarker-positive and biomarker-negative subgroups have different age distributions, the observed survival differences may partially reflect age effects rather than pure biomarker-treatment interactions. This analysis helps isolate the contribution of age heterogeneity to the overall results.

Limitations: This is a sensitivity analysis that removes natural age variation. The base case analysis using actual patient ages remains the primary analysis for decision-making.

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